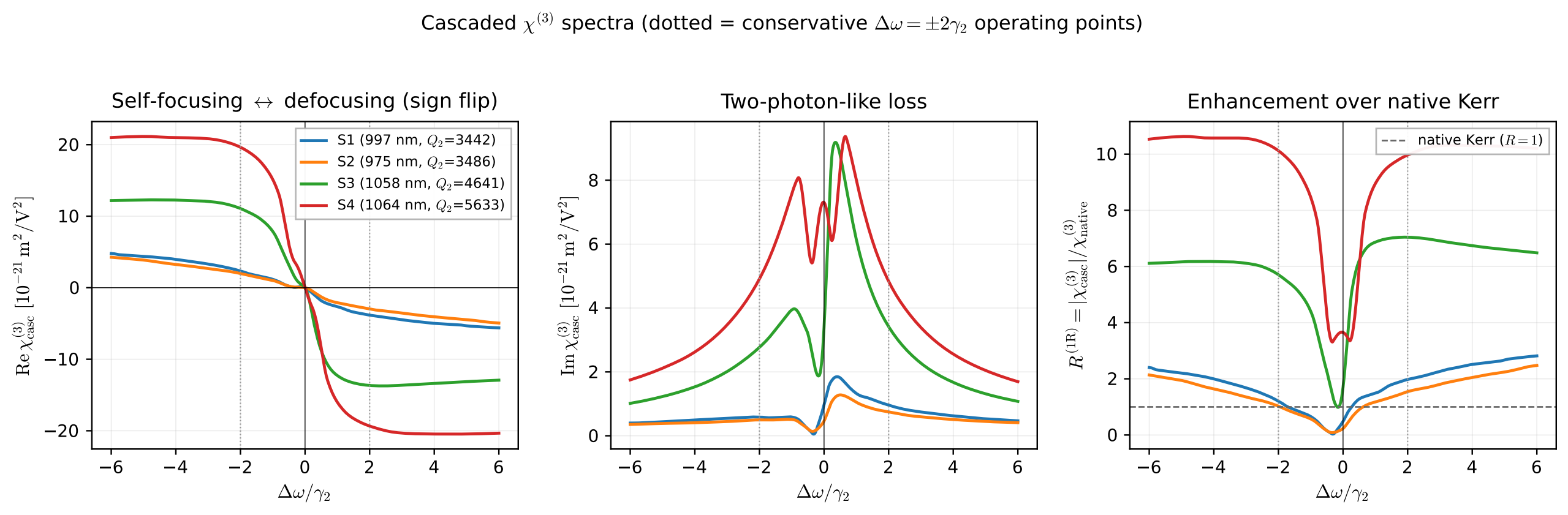
Cascaded $\chi^{(3)}$ spectra (dotted = conservative $\Delta\omega = \pm 2\gamma_2$ operating points)
Self-focusing ↔ defocusing (sign flip)
S1 (997 nm, Q_2=3442)
S2 (975 nm, Q_2=3486)
S3 (1058 nm, Q_2=4641)
S4 (1064 nm, Q_2=5633)
$\mathrm{Re}\,\chi^{(3)}_{\mathrm{casc}}$ [10^{-21} m^2/V^2]
$\Delta\omega/\gamma_2$
Two-photon-like loss
$\mathrm{Im}\,\chi^{(3)}_{\mathrm{casc}}$ [10^{-21} m^2/V^2]
$\Delta\omega/\gamma_2$
Enhancement over native Kerr
native Kerr ($R=1$)
$R^{(1R)} = |\chi^{(3)}_{\mathrm{casc}}|/\chi^{(3)}_{\mathrm{native}}$
$\Delta\omega/\gamma_2$